Data Flow Diagram

Date	15 March 2026
Team ID	xxxxxx
Project Name	AI based crop recommendation
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through the OptiCrop application between external entities, processes, and data stores. The DFD represents the flow of soil and environmental parameters from the farmer/user to the Machine Learning model and the final recommended crop result returned to the user.

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	Represents a person or system outside the application that sends or receives data (e.g., Farmer/User).
Rectangle with numbered header	**Process**	Represents activities that transform input data into output data (e.g., Input Collection, Data Preprocessing, ML Prediction, Result Display).
Rectangle (solid fill, no number)	**Data Store**	Represents stored information such as Crop Dataset, trained Machine Learning model, and scaler files.
Labeled arrow	**Data Flow**	Represents movement of information between entities, processes, and data stores (e.g., Soil Parameters, Prediction Result).

External Entity

Farmer / User

The user provides soil and environmental information and receives the recommended crop.

Processes

1.0 User Input Collection

- Collects soil and environmental parameters:
  - Nitrogen (N)
  - Phosphorus (P)
  - Potassium (K)
  - Temperature
  - Humidity
  - pH
  - Rainfall

2.0 Data Validation & Preprocessing

- Validates user inputs.
- Applies feature scaling using the trained scaler.

3.0 Machine Learning Prediction Engine

- Loads the trained model
- Processes the scaled input data.
- Predicts the most suitable crop.

4.0 Display Crop Recommendation

- Sends the predicted crop result back to the user through the Flask web interface.

Data Stores

D1: Crop Recommendation Dataset

- Stores crop data containing soil and environmental parameters.

D2: Trained Machine Learning Model

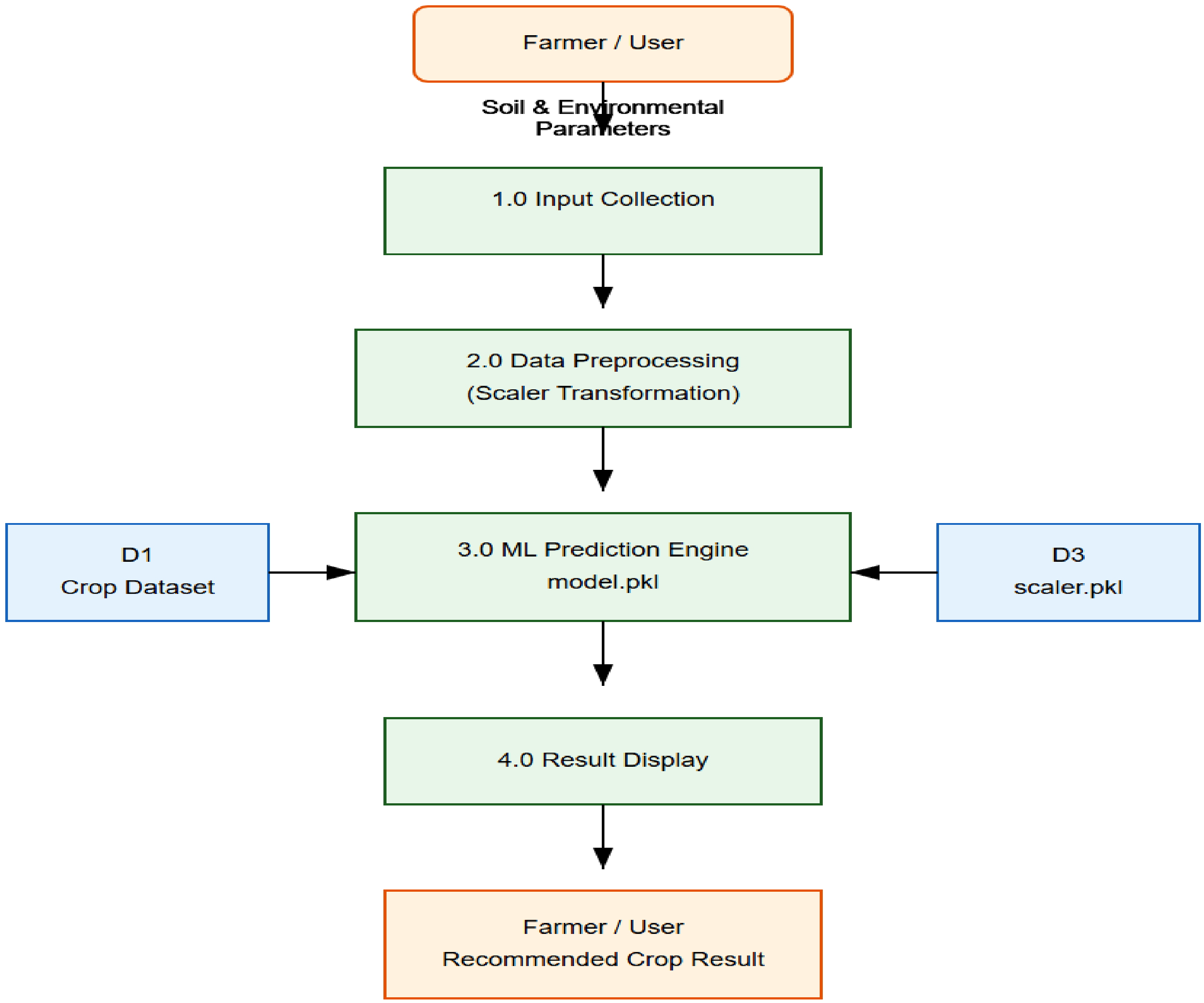
- Stores the trained prediction model used for crop recommendation.

D3: Feature Scaler

- Stores Standard Scaler used for preprocessing input values.

Data Flow

- Farmer/User → Soil & Environmental Parameters → Input Collection Process
- Input Collection → Validated Data → Preprocessing Process
- Preprocessing → Scaled Features → Machine Learning Model
- Machine Learning Model → Predicted Crop → Result Display
- Result Display → Recommended Crop → Farmer/User


Farmer / User
Soil & Environmental Parameters
1.0 Input Collection
2.0 Data Preprocessing
(Scaler Transformation)
D1
Crop Dataset
3.0 ML Prediction Engine
model.pkl
D3
scaler.pkl
4.0 Result Display
Farmer / User
Recommended Crop Result